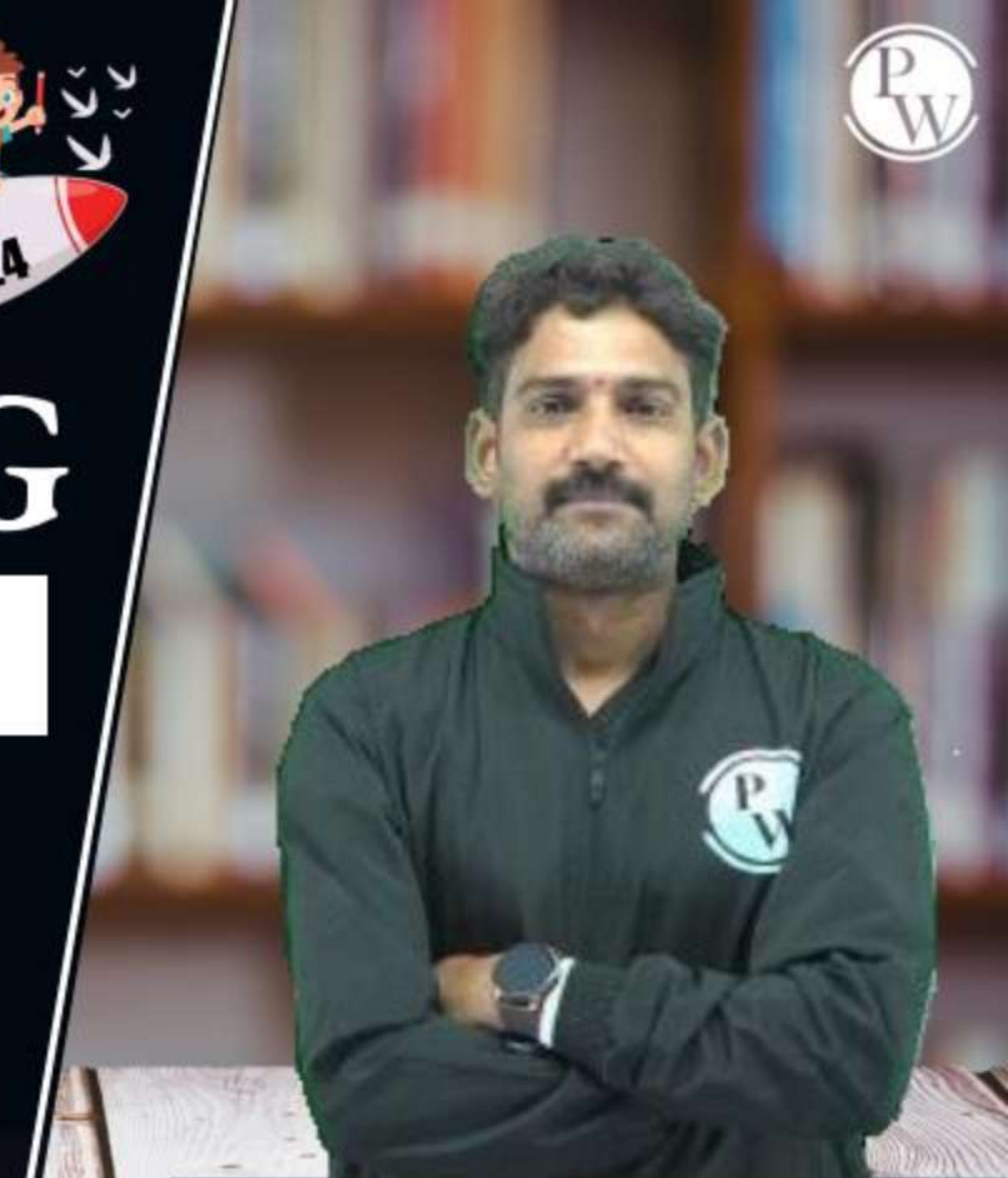
CS & IT ENGINEERING

'C' Programming

'C' TOKENS

Lecture No.- 04



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Recap of Previous Lecture

Unary operators

- Increment (++)
- Decrement (--)

Topics to be Covered

Binary operators

- Arithmetic
- Logical
- Bitwise and shift
- Relational
- Assignment

Topic : 'C' Tokens - 4

Arithmetic operators

+ (Addition), − (Subtraction), * (Multiplication), / (Division), % (Modulus)

Same as in Mathematics

/ (Division) : It Performs Numeric division and return Quotient as Result.

- The Result type depends on the input values.
  - If both inputs are integers, Result (Quotient) will also be integer.
  - Otherwise, Result also will be float value.

Quotient

divisor | Dividend

Remainder

Example

1) $a = 17,\ b = 3$

$c = a/b;$

$\boxed{c = 5}$

$$\begin{array}{r|l} & 5 = Q \\ 3 & 17 \\ & 15 \\ \hline & 2 = R \end{array}$$

2) $a = 5.5,\ b = 2$

$c = a/b;$

$c = 2.\underbrace{750000}_{\text{6-digit Precision}}$

$$\begin{array}{r|l} & 2.75 \\ 2 & 5.5 \\ & 4 \\ \hline & 15 \\ & 14 \\ \hline & 10 \\ & 10 \\ \hline & 0 \end{array}$$

3) $a = 5 \quad b = 2.0$

$c = a/b$

$c = 2.500000$

4) $a = 5.5 \quad b = 2.0$

$c = a/b$

$c = 2.750000$

5) $a = +7,\ b = -2$

$c = a/b$

$c = -3$

a sign	b sign	a/b sign
+	+	+
+	−	−
−	+	−
−	−	+

a sign	b sign	a % b sign
+	+	+
+	−	+
−	+	−
−	−	−

% (Modulus)

- It Performs Numeric [Integers] division, and Produces remainder as Result.
- For signed numbers, Numerator sign is assigned for result (Remainder)

Examples

$a = 7, \ b = 2$

$c = a \% b$

$\boxed{c = 1}$

$$\begin{array}{r} 3 \\ 2\overline{)7} \\ 6 \\ \hline 1 \end{array}$$

$a = +7 \quad b = -2$

$c = a \% b$

$\boxed{c = 1}$

$a = -7, \ b = 2$

$c = a \% b$

$\boxed{c = -1}$

$a = -7 \quad b = -2$

$c = a \% b$

$\boxed{c = -1}$

$a = 7.0 \quad b = 2.0$

$c = a \% b$

$\Rightarrow$ Error, modulus Cannot be used on float values.

Binary operators — Unary operator

Logical operators : Logical AND (&&), Logical OR (||), ! (Logical NOT)

- These Perform operation on Truth Values (True / False)

Number 0, '\0' (NULL character) ⇒ False

Other (Non-zero) ⇒ True

A	B	A && B	A || B	!A	!B
F	F	F	F	T	T
F	T	F	T	T	F
T	F	F	T	F	T
T	T	T	T	F	F

Examples

1) $a = 7,\ b = 5,\ c = 12$

$d = a > b$ $d = $ True / 1

$e = c < a$ $e = $ False / 0

$f = a \mathrel{!}= b$ $f = $ True / 1

$g = b <= c$ $g = $ True / 1

2) $a = 7,\ b = 0,\ c = -3,\ d = 4.5$

$f = $ '#', $g = $ '\0'

$x = a \,\&\&\, b \Rightarrow 7 \,\&\&\, 0 \Rightarrow T \,\&\&\, F = F = 0 \Rightarrow x = 0$

$y = b \,||\, c \Rightarrow 0 \,||\, -3 \Rightarrow F \,||\, T = T \Rightarrow 1 \Rightarrow y = 1$

$z = f \,||\, g \Rightarrow T \,||\, F \Rightarrow T \Rightarrow 1 \Rightarrow z = 1$

Short-Circuiting

Logical AND ($\&\&$) $\Rightarrow$ when First input is 'False' $\Rightarrow$ Second input is Not Evaluated

Logical OR ($||$) $\Rightarrow$ when First input is 'True' $\Rightarrow$ Second input is Not Evaluated.

a	b	c
$\cancel{5}\ \cancel{6}\ 5$	$\cancel{0}\ \cancel{1}\ 0$	-3

Examples

1) $a = 5,\ b = 0,\ c = -3$

$x = ++a \ \&\&\ b++$ $\quad 6 \ \&\&\ 0$ $\quad$ True && False $\Rightarrow$ False/0

$y = --b \ \&\&\ \cancel{c++}$ $\quad 0 \ \&\&\ \_$ $\quad$ False $\Rightarrow y = 0$

$z = a-- \ ||\ \cancel{++c}$ $\quad 6 \ ||\ \_$ $\quad$ True $\Rightarrow z = 1$

$a = \underline{5}$ $\quad b = \underline{0}$ $\quad c = \underline{-3}$ $\quad x = \underline{0}$ $\quad y = \underline{0}$ $\quad z = \underline{1}$

p	q	r	s	t
~~7~~ ~~8~~ 7	~~0~~ ~~1~~ ~~0~~ 1	~~-3~~ -4	~~5~~ ~~4~~ ~~5~~ ~~4~~ 5	0 1

2) $p=7,\ q=0,\ r=-3,\ s=5,\ t=0$

$(\&\&) > (||)$

$a = p \&\& s \quad 7 \&\& 5 \Rightarrow T \&\& T = T/1$

$b = q \,||\, r \quad 0 \,||\, -3 \Rightarrow F \,||\, T = T/1$

$c = q{+}{+} \,||\, {+}{+}t \quad 0 \,||\, 1 \Rightarrow F \,||\, T = T/1$

$d = --s \&\& --q \quad 4 \&\& 0 \Rightarrow T \&\& F = F/0$

$e = \boxed{p{+}{+} \,||\, \boxed{r{-}{-} \&\& {+}{+}s}} \,||\, \cancel{-q}$

True

$-3 \&\& 5 = T \&\& T = \text{True}$

$7 \,||\, \_\_ = \text{True}$

$T \,||\, \_\_ = \text{True} \Rightarrow e=1$

$f = \underbrace{\boxed{s{-}{-} \&\& p{-}{-}}}_{T} \,||\, \underbrace{\boxed{s{+}{+} \&\& q{+}{+}}}_{F}$

$= \text{True}/1$

$5 \&\& 8 = \text{True} \&\& \text{True} = \text{True}$

$4 \&\& 0 \Rightarrow T \&\& F = F$

$\underbrace{\boxed{\boxed{s{-}{-} \&\& p{-}{-}} \,||\, s{+}{+}}}_{\text{True}} \,||\, q{+}{+}$

short circuited

$\text{True} \,||\, q{+}{+}$

short circuited

$p=7$
$q=1$
$r=-4$
$s=5$
$t=1$
$a=1$
$b=1$
$c=1$
$d=0$
$e=1$
$f=1$

Bitwise and shift operators

`&` (Bitwise AND), `¦` (Bitwise OR), $\underbrace{\sim \text{(Bitwise NOT)}}_{\text{unary}}$, `^` (Bitwise XOR), `<<` (Left shift), `>>` (Right shift)

- Bitwise and shift operators Perform operation on binary digits (bits = 0/1)

A	B	A & B	A ¦ B	~A	~B	A ^ B
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	1	0	0	0

[Bit wise operations Performed with standard bit Representations ⇒ 8 bits | 16 bits | 32 bits ..]

Examples

1) $a = 17 \quad b = 12$

$c = a \& b \Rightarrow 17 \& 12 \Rightarrow c = 0$

$d = a \mid b \Rightarrow d = 29$

$e = a \wedge b \Rightarrow e = 29$

$(17)_{10} = 00010001$

$(12)_{10} = 00001100$

$17 \& 12 = 00000000 \Rightarrow 0$

$a = 17 = 0001\ 0001$

$b = 12 = 0000\ 1100$

$17 \mid 12 = 0001\ 1101 \Rightarrow 29$

7	6	5	4	3	2	1	0
0	0	0	1	1	1	0	1

$1\times2^0 + 0\times2^1 + 1\times2^2 + 1\times2^3 + 1\times2^4$
$+ 0\times2^5 + 0\times2^6 + 0\times2^7 = 1+4+8+16 = 29$

$a = 17 = 0001\ 0001$

$b = 12 = 0000\ 1100$

$17 \wedge 12 = 0001\ 1101$

2	17	LSB
2	8	→ 1
2	4	→ 0
2	2	→ 0
2	1	→ 0
	0	→ 1
		MSB

2	12	
2	6	→ 0
2	3	→ 0
2	1	→ 1
	0	→ 1

Shift operators(<<, >>)

$$\boxed{\text{Value} << \text{Count} \equiv \text{Value} * 2^{\text{Count}}}$$

$$\boxed{\text{value} >> \text{Count} \equiv \text{Value} / 2^{\text{Count}}}$$

Syntax:

$$\boxed{\begin{array}{l} \text{Value} << \text{Count} \\ \text{Value} >> \text{count} \end{array}}$$

Ex:

$a << b$

$x >> 3$

Ex:

$a = 14$

$b = a << 3$

$b = 14 << 3$

$= 14 * 2^3$

$= 14 * 8$

$\boxed{b = 112}$

$(14)_{10} = 0000\,1110 \Rightarrow 14$

<< 1 $\quad$ $\cancel{0}$ $\quad$ 0001 1100 $\Rightarrow 28$ $\quad$ *2

<< 1 $\quad$ $\cancel{0}$ $\quad$ 0011 1000 $\Rightarrow 56$ $\quad$ *2

<< 1 $\quad$ 0 $\quad$ 0111 0000 $\Rightarrow 112$ $\quad$ *2

$\quad$ 0 $\quad$ 1110 0000 $\Rightarrow 224$ $\quad$ *2

00000000 11100000

Ex:

$a = 14$

$b = a >> 2$

$b = 14 >> 2$

$= 14 / 2^2$

$= 14 / 4$

$b = 3$

14 $\Rightarrow$ 0000 1110

/2 $\quad$ >> 1 $\quad$ $\cancel{0}$

7 $\Rightarrow$ 0000 0111

/2 $\quad$ >> 1 $\quad$ $\cancel{1}$

3 $\Rightarrow$ 0000 0011

Relational (or) Comparison Operators

- $<, <=, >, >=, ==$ (Equal to), $!=$ (Not Equal to)
- Thes Produce result in Truth values (T / F)

Ex: $a=4,\ b=5,\ c=3$

$x = a < b \Rightarrow x = \text{True} \mid 1$

$y = c > b \Rightarrow y = \text{False} \mid 0$

$z = a < b > c$

~~$a<b$ and $b>c$~~

~~$4<5$ and $5>3$~~

~~T and T = True=1~~

$(a<b) > c$

$(4<5) > 3$

$\text{True}\mid 1 > 3 \Rightarrow \text{False} = 0$

$\boxed{z=0}$

Assignment operator (=)

$==$ Vs $=$

$==$	$=$
compare 2 values and return True/1 if both are same, otherwise False/0	copies RHS value into LHS.

Ex: $a=7,\ b=5$

$x = a == b \Rightarrow x = \text{False} \Rightarrow x=0$

$a=7,\ b=5$

$x = a = b$

$x=5 \quad a=5 \quad b=5$

2 mins Summary

Binary operators

- Arithmetic
- Logical $\longrightarrow$ Short Circuiting
- Bitwise and shift
- Relational
- Assignment

To be Contd . . .

THANK - YOU